MIL-F-0011137E(EA)
1 December 1987
SUPERSEDING
MIL-F-0011137D(EA)
3 February 1982
USED IN LIEU OF
MIL-F-11137C
30 August 1963

MILITARY SPECIFICATION

FILTER, GAS, 150 CFM, M10

This military specification has been prepared by the U.S. Army Chemical Research, Development and Engineering Center (SMCCR-SPT-S) based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-F-11137C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in aquisition.

1. SCOPE

This specification covers a gas filter for use in removing toxic gases from air forced through the filter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

DISTRIBUTION STATEMENT A. Approved for public release; distribution *is* unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:28Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.

MILITARY

MIL-C-13724 - Charcoal, Activated, Impregnated, ASC.
MIL-F-46032 - Fungus-Resistant Treatment (for Fabrics) Copper Processes,

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Test Procedures for Packaging Materials.
FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes,
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance-Test Methods.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-19-673 - Filter, Gas, 150 cfm, M10.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-673 - Filter, Gas, 150 cfm, M10.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:28Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the *Issues* of the documents which are DOD adopted shall be those listed In the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

ASTM STANDARDS

D 2867 - Moisture Inactivated Carbon, Test Methods For.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are *normally* available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-673 *or* on the subsidiary drawings shall conform to the specifications listed thereon *or to* the specific characteristics set forth on the drawing.

3.1.1.1 Cotton cloth.

(a) Fungus resistance. The fines filter cotton cloth shall be made fungus resistant in accordance with MIL-F-46032. Processing shall be type III, method optional, except that leaching shall be omitted. No visible growth of fungus shall be evident *when* tested in accordance with method 5750 of F'ED-STD-191 l

(b) Water repellency. The cotton cloth shall have a minimum spray rating of 50.0 when tested in accordance with method 5526 of FED-STD-191.

3.1.2 Components. All components of the gas filter shall conform to the specifications and drawings listed on Drawing 5-19-673 and on the subsidiary drawings.

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T15:28Z
Check the source to verify that this is the currentversion before use.

3.2 Assembly. The gas filter shall be assembled as specified on Drawing 5-19-673. During the assembly operation, the moisture content of the charcoal shall not exceed 2.0 percent when tested as specified in 4.4.4.l.

3.3 Performance.

3.3.1 Airflow resistance. The airflow resistance of the gas filters shall be 4.5 ± 0.5 inches of water at an airflow rate of 150 cfm ± 10 cfm at 75° ± 5°F and 30 percent maximum relative humidity when tested as specified in 4.4.4.2.

3.3.2 Leakage. The gas filter shall not leak when a concentration of 1000 parts per million (ppm) of dichlorodifluoromethane refrigerant (R-12), conforming to type 12 of BB-F-1421, in air at 75° ± 5°F and 30 percent maximum relative humidity is introduced at the influent side of the filter at one-fifth the' rated airflow of the filter when tested as specified in 4.4.4.3. A filter leak shall be defined as the detection of 1 ppm or more of R-12 at the effluent side of' the filter within 2 minutes after the introduction of R-12 at the influent side.

3.3.3 Airflow resistance and dimethylmethylphosphonate (DMMP) value after rough handling (destructive) . The gas filter shall be rough handled in the upright position for 15 minutes in accordance with 4.4.4.7. After rough handling, the gas filter shall show no evidence of cracks, dents, or charcoal leakage; the airflow resistance shall comply with 3.3.1; and when tested as specified in 4.4.4.4, the filter shall not permit penetration of 0.04 micrograms or more of DMMP per liter of air within a minimum of 75 minutes.

3.3.L CK gas life of charcoal sorbent. After completion of all manufacturing steps, the filter's charcoal shall comply with cyanogen chloride (CK) gas sorption requirement of MIL-C-13724 for grade IV unaged charcoal when tested in accordance with 4.4.4.6.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of gas filters shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The gas filter shall be free from chipped, burred, or bent metal sections, ripped or torn cloth, abraded gaskets, and foreign matter (dirt, oily or viscous material).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T15:28Z
Check the source to verify that this is the currentversion before use.

right to perform any of the inspections set forth In the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not *relieve* the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 20 gas filters (see 6.2(d)) shall be produced using the same materials, methods, and equipment as will be used in regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Each gas filter of the preproduction lot shall be examined for all the requirements of this specification.

4.3.2.2 For nondestructive tests.

4.3.2.2.1 Moisture content of charcoal sorbent. The sample for moisture content shall be taken at the station where the filter is filled with sorbent. A sample specimen of charcoal sorbent shall be taken during the filling of the first, tenth, and twentieth filter in the production lots. Each sample (25 grams) shall be sequentially designated and suitably identified as from the preproduction lot.

4.3.2.3 For destructive test. Eight filters shall be tested for airflow resistance and DMMP life after rough handling, and four filters shall be tested for CK life of charcoal sorbent.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:28Z
Check the source to verify that this is the currentversion before use.

4.3.3 Inspection procedures.

4.3.3.1 For examination. Each gas filter in the preproduction lot shall be examined for the defects listed in the classification of defects. If any filter is found defective, the preproduction lot shall be rejected.

4.3.3.2 For nondestructive tests.

(a} Moisture content of charcoal sorbent. The sample specimen shall be tested for moisture content in accordance with 4.4.4.1. If any specimen fails to meet the requirement for moisture content, the preproduction lot shall be rejected.

(b) Airflow resistance and leakage. Erich gas filter in the preproduction lot shall be tested for airflow resistance In accordance with 4.4.4.2, and leakage in accordance with 4.4.4.3. If any filter fails to meet the requirement for airflow resistance or leakage, the preproduction lot shall be rejected.

4.3.3.3 For destructive test.

(a) Airflow resistance and DMMP gas life after rough handling. The sample filters from the preproduction lot shall be tested for airflow *resis*-tance and DMMP gas life after being rough handled in accordance with 4.4.4.7 (see 3.3.3). If any sample filter fails to meet the requirement for airflow resistance or DMMP gas life after rough handling, the preproduction lot shall be rejected. Each sample filter shall meet the minimum DMMP gas life requirement.

(b) CK gas life of charcoal sorbent. The sample filters shall be tested in accordance with and meet the test requirements of 4.4.4.6.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with all the requirements of this specification when examined and tested in accordance with 4.3.3 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the gas filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of charcoal sorbent shall be represented in any one lot of finished gas filters.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted In accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T15:28Z
Check the source to verify that this is the currentversion before use.

4.4.2.2 For nondestructive tests.

4.4.2 .2.1 Moisture content of charcoal sorbent. The sample taken from the station where the filter is filled with sorbent shall be designated as follows:

(1) A sample taken during the filling of the first filter of each day.

(2) A sample taken during the filling of the last filter of each day or whenever production is interrupted or a new lot of sorbent is introduced.

A sample for moisture content (25 grams) shall be designated and suitably identified for each lot of gas filters produced.

4.4.2.3 *For* destructive tests.

(a) Airflow resistance and DMMP gas life after rough handling. Sampling shall be in accordance with MIL-STD-414, level I, AQL 1.0 percent.

(b) CK gas life of charcoal sorbent. Four filters shall be tested for gas life.

4.4.3 Inspection procedures.

4.4.3.l For examination. Sample gas filters shall be examined in accordance with the classification of defects (4.4.3.4) and MIL-STD-105.

4.4.3.2 *For* tests.

(a) Moisture content of charcoal sorbent. The sample specimen shall be tested for moisture content in accordance with 4.4.4.1. If any specimen fails to meet the requirement for moisture content, the gas filters represented by the sample specimen shall be rejected.

(b) Airflow resistance and DMMP gas life value after rough handling. The sample gas filters shall be tested for airflow resistance and DMMP gas life value after rough handling in accordance with 4.4.4.4. If any sample filter fails to meet the requirement for airflow resistance or shows evidence of damage (see 3.3.3) after rough handling, the lot represented by the sample shall be rejected. All sample filters shall meet the minimum DMMP gas life.

(c) CK life of charcoal sorbent. The sample filters shall be tested as specified in 4.4.4.6.

4,4.3.3 For critical defects. Each gas filter in the lot shall be inspected for the critical characteristics listed in the classification of defects. If any filter fails to meet the requirements of 3.3.1 or 3.3.2, only that filter shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:28Z
Check the source to verify that this is the currentversion before use.

4.4.3.4 Classification of defects.

(a) Filter, gas, 150 CFM, M10 (Dwg 5-19-673).

Categories	Defects	Acceptance standards
Critical:		
1	Airflow resistance	4.4.4.2
2	Leakage	4.4*4*3
Minor:	AQL 1.0 percent defective	
101	Overall dimensions	
102	Component missing	
103	Component incorrectly assembled	
104	Workmanship (3.5)	
105	Marking incorrect or illegible	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing	

(b) Frame (Dwg 5-19-669).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Welding, as applicable
102	Flatness and squareness

(c) Packaging.

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Item not completely cleaned/dried	
102	Interior container missing or not as specified	
103	Unit, intermediate, or shipping container not as specified	
104	Closure of unit, intermediate, or shipping container not as specified	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:28Z
Check the source to verify that this is the current version before use.

105	Marking of unit, intermediate, or shipping container missing, incorrect, or illegible	
106	Unit container *barrier leakage*	4.4.4.5

4.4.4 Tests. Tests shall be conducted as follows:

4,4.4.1 Moisture content of charcoal sorbent. Moisture content of the charcoal sorbent shall be determined according to ASTM D 2867.

4.4.4.2 Airflow resistance. Insert the gas filter into a test apparatus. Regulate the air source to the influent side of the filter to the specified airflow rate. Record the resistance as indicated by the manometer.

4.4.4.3 Leakage. Test for R-12 as follows:

(a) Connect the filter to a source of forced air.

(b) Regulate the flow of air from the blower to 30 + 2 scfm.

(c) Introduce R-12 into the intake of the blower. Monitor and maintain a concentration of 1000 + 20 ppm by volume of R-12 in air on the influent side of the filter for a minimum of 2 minutes. Do not exceed 3 minutes airflow.

(d) Determine the leakage using a leak detector capable of detecting 1 ppm of R-12.

(e) Purge the filter by passing fresh air in the reverse direction through the filter at its rated flow for 1 minute.

4.4.4.4 Dimethylmethylphosphonate (DMMP) value after rough handling. After the filter has been rough handled in the vertical position for 15 minutes as specified in 4.4.4.7, the sample filter shall be tested for DMMP value in an apparatus ‘which establishes the following test conditions:

(a) Preheat sample filters to 110 + 5°F in a forced draft oven for 30 minutes.

(b) Insert an expendable filter, not a sample filter for test, in the test apparatus and start the blower.

(C) Adjust the controls until the rated airflow is reached and the temperature is 110 ± 5°F.

(d) Generate DMMP until the influent concentration detector indicated 5.0 + 0.25 milligrams DMMP per liter air. Note the generator and systems settings.

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T15:28Z
Check the source to verify that this is the currentversion before use.

sufficient capacity to maintain a cooling liquid

(e) Shut down the tester, remove the expendable filter, and insert the preheated sample filter to be tested.

(f) Start the tester without DMMP and quickly bring the tester to the proper flow, temperature, and relative humidity.

(g) Start the flow of DMMP and quickly bring the concentration to - 5 + 0.25 milligrams DMMP per liter air.

(h) Start the timer when the influent concentration detector first indicates DMP in the systems.

(i) Stop the test and record the time when the effluent concentration detector indicated a concentration of 0.04 micrograms per liter air.

4.4.4.5 Barrier leakage. Twelve hours after sealing of filter packaged in barrier bag, inspect the package visually. The barrier bag should be in contact with all surfaces of the inner box.

4.4.4.6 CK gas life of charcoal sorbent. Disassemble the filters. A 250 ± 5 milliliter sample of charcoal shall be removed from each filter and placed in a separate canister. The samples shall be tested for CK gas sorption time in accordance with MIL-C-13724. Check the average sorption time for compliance with the criteria of MIL-C-13724.

4.4.4.7 Rough handling. Unpackaged filter shall be rough handled in accordance with test method T-105.10 of MIL-STD-282. No filter shall be subjected to rough handling more than once regardless of the number of tests subsequently performed *on* the filter.

4,4.4.8 Heat seal seam strength. The heat seal seam of the unit pack container shall show no sign of heat seam separation when tested in accordance with Fed. Test Method Std. No. 101, method 2024, except that the static load shall be 50 ounces + 2 ounces, and 5 percent reduction in the static load shall be permitted when room temperature in the test area exceeds 90°F.

5. PACKAGING

5.1 Packaging. The gas filters shall be packaged in accordance with Special Packaging Instruction (SPI) P5-19-673.

5. NOTES

6.1 Intended use. The gas filter covered by this specification is primarily intended for use with the ABCA-M6A1 filter units.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction:

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:28Z
Check the source to verify that this is the current version before use.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production

(c) Level of packing required (see section 5).

(d) Disposition of preproduction filters from acceptable lots. The preproduction filters which do not experience destructive tests may be considered deliverable items. If accepted for delivery, the time and destination of shipment shall be specified.

6.3 Filling procedures. Filters filled in the following manner have been found to satisfy the performance requirements of this specification.

(a) Place a filter frame with the bottom perforated sheet, fine filters, and draw-bolt assembly in a position for filling in a suitable gravity type filling machine.

(b) Feed a measured amount of properly screened sorbent gradually and uniformly from a three-foot height onto the filter frame by means of the filling machine. The weight of sorbent required to properly fill the frame will vary according to its bulk density and moisture content. The required weight has been found to be approximately 28 pounds per frame. It is advisable to predetermine experimentally the required amount of sorbent to fill a frame.

(c) After filling, place the frame on a table and carefully brush the surface sorbent back from the edge on each of the four sides to about two inches from the outer edges so as to form a bed level 18 inches square, elevated 1/4 to 3/8 inch above the aluminum frame.

(d) Position the fines filter, perforated sheet, CUP washer gasket, and cup washer on the draw bolt with the edges of the fines filter and perforated sheet properly aligned with the filter frame. Assemble the washer and nut to the draw-bolt, tightening the nut to an approximate torque of 40 inch-pounds. Clamp the edges of the perforated sheet to the aluminum frame with a suitable mechanical press and fasten the perforated sheets in place with thread cutting tapping screws.

(e) Remove the assembly from the press and tighten the draw bolt to an approximate torque of 85 inch-pounds. The draw bolt may require a greater or lesser torque. The nut should be tightened to the highest torque that will not cause splitting of the perforated sheet at the time of tightening or when the filter is rough handled.

(f) Complete the assembly as shown on Drawing 5-19-673.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:28Z
Check the source to verify that this is the current version before use.

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 296. When amendment , revision, or cancellation of the specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.5 Interplant shipment. Packaging for supplies and materials which will not enter the military supply system. Typical are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor or from a vendor or contractor to a military arsenal or plant.

6.6 Subject term (key word) listing.

Air purification
Airflow resistance
Charcoal sorbent
Collective protection equipment
Filter, gas
Filter gas life
Toxic gas removal

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army-EA

Preparing activity:

Army-EA

Project No. 4240-A921

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

Commander
U.S. Army Chemical Research, Development
and Engineering Center
ATTN: SMCCR-SPT-S
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 202610-17T15:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-0011137E(EA)	FILTER, GAS, 150 CFM, M10

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 20261O-17T15:28Z
Check the source to verify that this is the current version before use.